Practice Problem Set 4

ECON 101 – Macroeconomics

Fall 2025

1. Explain the three main sources of long-run economic growth. Then, using Canada as an example, describe one policy that could enhance each source.

2. Suppose the production function of a small open economy is:

$$Y = AK^{0.3}L^{0.7}$$

where A is total factor productivity (TFP). If $A = 5$, $K = 400$, and $L = 100$:

(a) Calculate the level of real GDP.

(b) If capital increases by 10% and labour remains constant, what is the percentage increase in output? (Use the exponent rule.)

(c) Explain why output increases by less than 10%.

3. A country's real GDP grows by 4%. Capital grows by 2%, labour by 1%, and the capital and labour shares are 0.4 and 0.6 respectively.

(a) Compute the contribution of capital, labour, and total factor productivity (TFP) to growth.

(b) Interpret what a negative TFP growth would imply for an economy.

4. Discuss how the following policies can affect long-run growth:

(a) Increasing research and development (R&D) subsidies

(b) Expanding higher education

(c) Strengthening property rights and reducing corruption

5. Write down the components of aggregate expenditure (AE). Which component is the largest in the Canadian economy? Explain how *planned investment* differs from *actual investment*.

6. Consider the following data for a closed economy (all values in billions of dollars):

Component	**Function / Value**
$C = 100 + 0.8Y_D$	where $Y_D = Y - T$
$I = 150$	
$G = 200$	
$T = 100$	

(a) Write the aggregate expenditure function $AE = C + I + G$.

(b) Solve for equilibrium real GDP ($Y = AE$).

(c) What is the marginal propensity to consume (MPC)?

(d) Calculate the multiplier.

(e) Suppose government spending increases by $50 billion. Compute the new equilibrium GDP.

7. Explain what happens to unplanned inventories when:

(a) Actual output > planned AE

(b) Actual output < planned AE

How do firms respond in each case to return the economy to equilibrium?

8. Suppose net exports are given by:

$$NX = 50 - 0.1Y$$

Using the same parameters from Question 6, find the new equilibrium level of GDP. Explain intuitively how the *marginal propensity to import* affects the multiplier.